

Spatial Analysis of Gene Expression Patterns in Human Lung Cancer

A Comparative Study of GraphST and BANKSY

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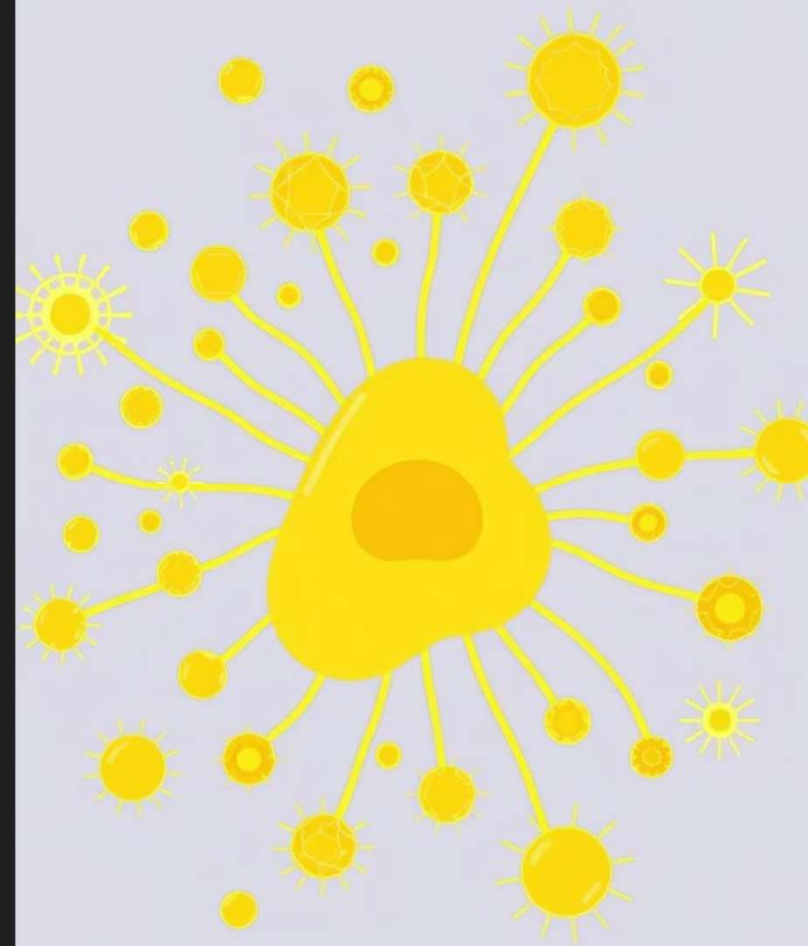
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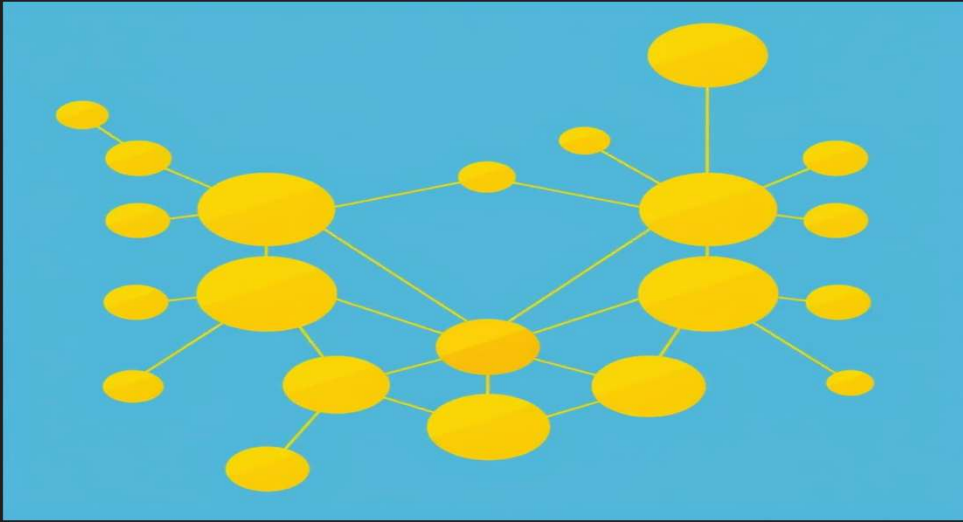
Motivation: Why AI for Spatial Analysis in Lung Cancer?

The spatial arrangement of the **Tumour Microenvironment (TME)** is critical for lung cancer development, metastasis, and therapy response.

Spatial Transcriptomics (ST) is a key technology for mapping the TME's complex architecture at a molecular level. ST experiments produce massive datasets (tens of thousands of coordinates, thousands of genes), making AI-driven computational models essential to identify meaningful biological patterns.

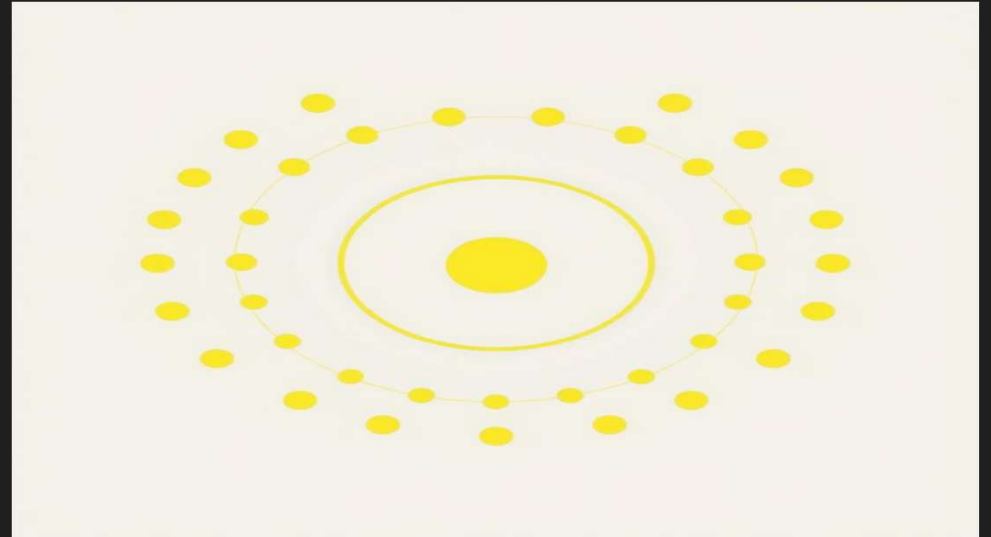


Two Competing AI Philosophies: GraphST vs. BANKSY



GraphST (Global Approach)

- Uses a **Graph Neural Network (GNN)** to learn a combined embedding of spatial and transcriptional similarity.
- Designed to detect spatially continuous domains and maintain tissue integrity.



BANKSY (Local Approach)

- Uses a **filter/kernel-based framework** to interpret local neighborhoods.
- Designed to capture subtle expression transitions and local heterogeneity.

Project Contribution: The first head-to-head comparison of GraphST and BANKSY on a Visium HD lung cancer ROI.

Methodology: A Fair & Reproducible Comparison



This rigorous methodology minimizes confounding variables, providing unprecedented clarity in comparing these two distinct AI approaches.

- **Single Shared ROI:** Both methods analyzed the exact same tissue region: 14,880 spots and over 18,000 genes.
- **Harmonised Pipeline:** Identical preprocessing, clustering (Leiden), and marker discovery (Wilcoxon test) were used for both branches.
- **Deterministic Execution:** Fixed seeds and versioned code ensure 100% reproducibility of all artifacts.

Key Finding 1: Different Views, Same Geography

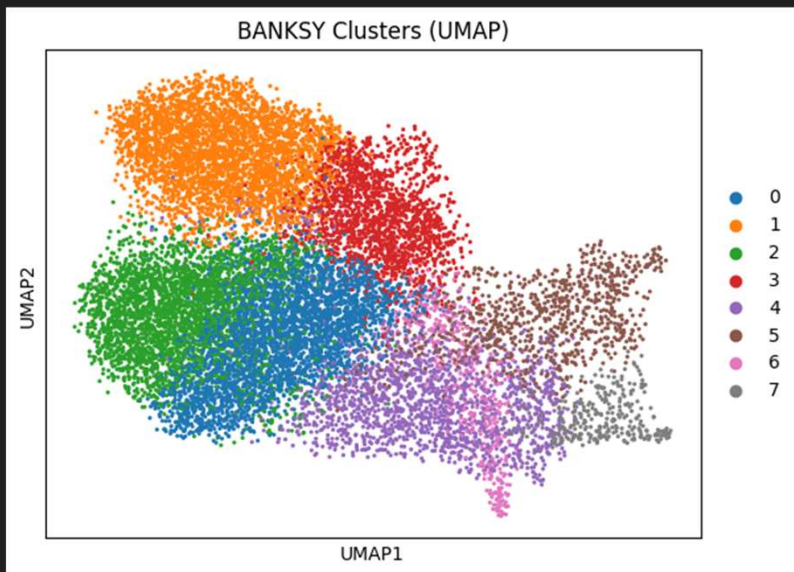


Fig: Banksy UMAP

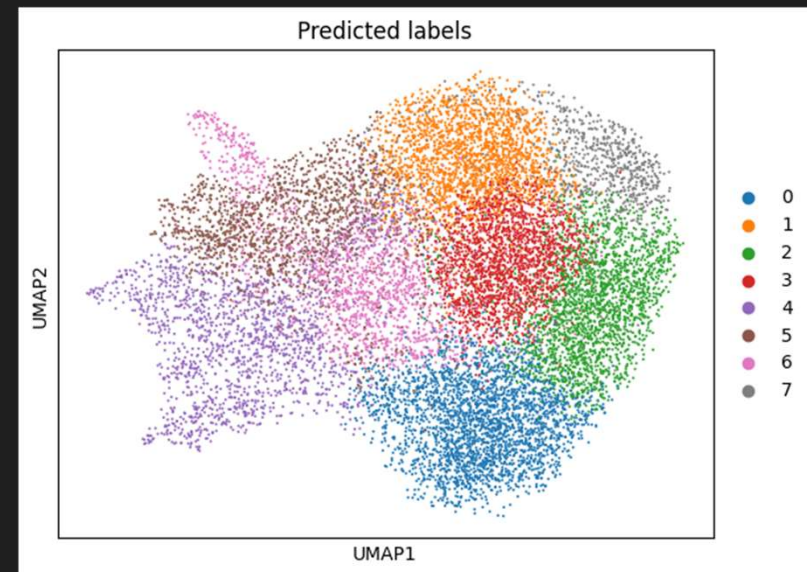
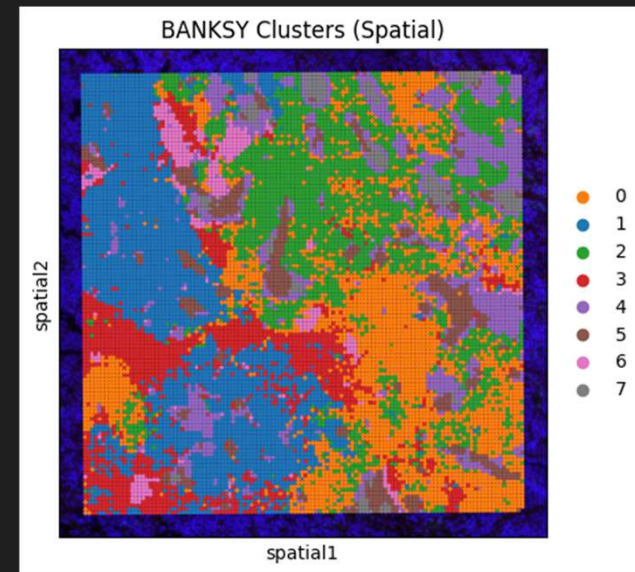
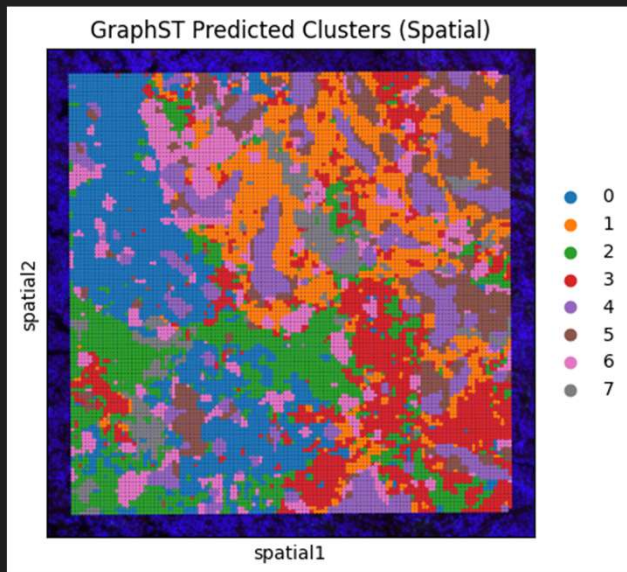


Fig: GraphST UMAP

- **Different Learned Geometry:** GraphST preserves smooth chains, suggesting gradual transitions, while BANKSY emphasizes crisp island separation.
- **Consistent Spatial Geography:** Despite different UMAPs, both models form large, coherent domains in comparable physical locations on the tissue slide.
- **Conclusion:** Both methods find robust, real biological structures; their main difference lies in how they represent the *boundaries* between them.

Spatial Cluster Maps: GraphST vs. BANKSY



Both GraphST and BANKSY models demonstrate successful identification of large, coherent tissue domains within the lung cancer Region of Interest (ROI), indicating robust detection of primary tissue compartments.

A crucial distinction emerges in how these models delineate cluster boundaries:

- **GraphST:** Generates more transitional belts at interfaces, a characteristic reflective of its graph-learning methodology that accommodates gradual state changes between spatial domains.
- **BANKSY:** Emphasizes crisp, well-defined borders and exhibits fewer "salt-and-pepper" artifacts, consistent with its approach of leveraging neighbor-enriched features to enhance local contrast.

Despite these methodological differences in boundary definition, the first-order adjacencies—that is, the qualitative neighboring relationships between clusters—are largely conserved across both models, suggesting a shared understanding of macro-level tissue organization.

Key Finding 2: Highlighting Interfaces vs. Coherent Domains

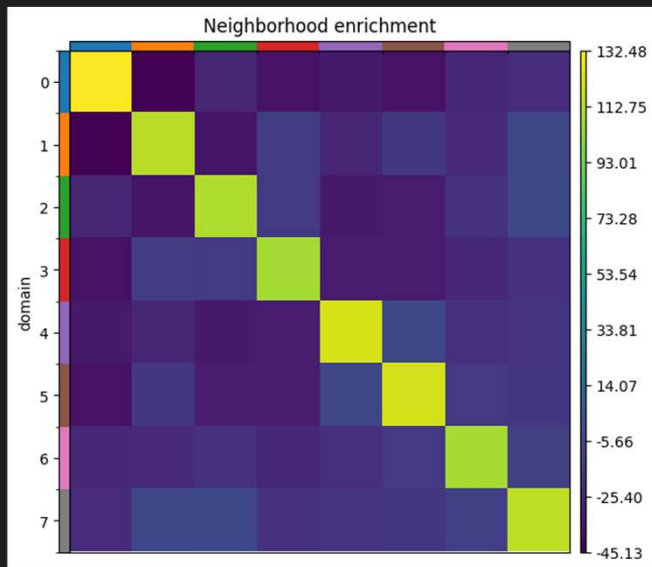


Fig: GraphSt Enrichment

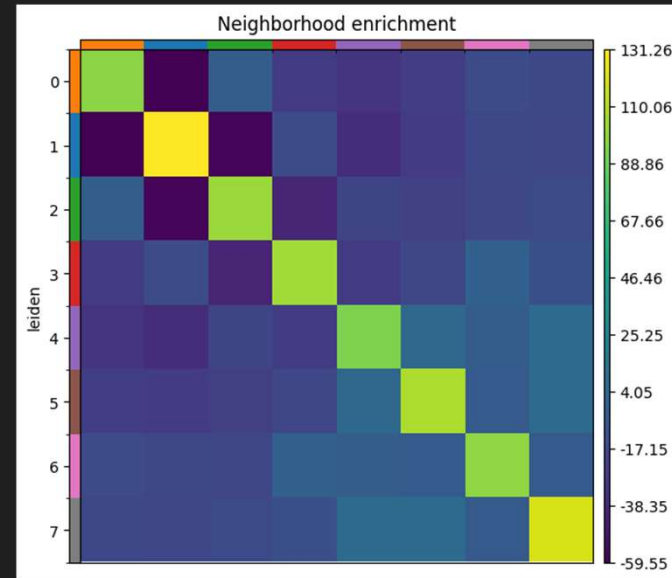


Fig: Banksy Enrichment

- **Neighbourhood Enrichment** quantifies spatial contact patterns between cluster pairs.
- **Bright Off-Diagonals** indicate enriched interfaces between different clusters.
- **Key Difference:** BANKSY shows slightly stronger off-diagonal contrast, better highlighting local boundaries, while GraphST's brighter diagonal reflects its focus on globally smoother domains.

Functional Annotation (GO/Enrichr): Broader vs. Specialised

Gene Ontology (GO) was used to assign biological functions to each cluster's marker genes.

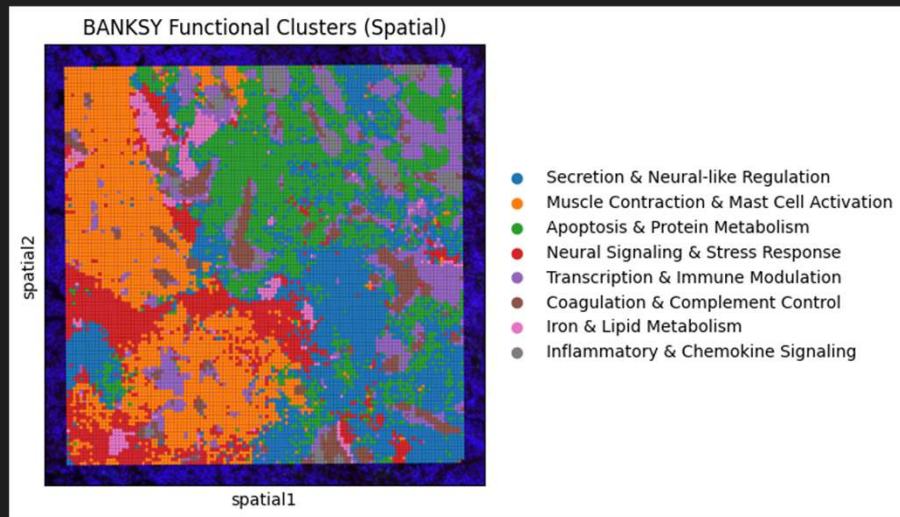


Fig: BANKSY Go Analysis

BANKSY: Its boundary-centric features lead to more specialized labels at interfaces, such as "**Inflammatory & Chemokine Signaling**".

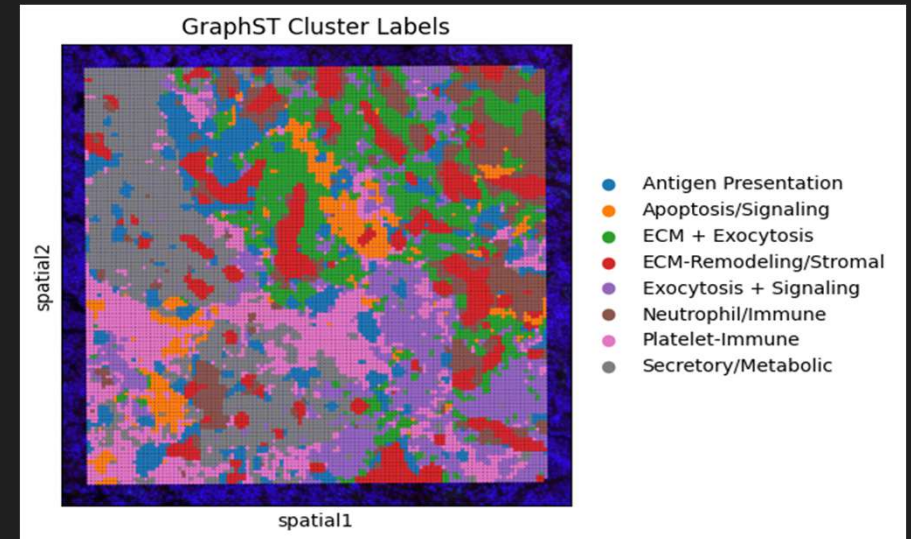


Fig: GraphSt Go Analysis

GraphST: Its emphasis on smooth transitions results in broader functional domains, like "**ECM-Remodeling/Stromal**".

Cell-Type Annotation (SingleR): Mixed vs. Niche-Specific

SingleR was used to map clusters to likely cell types by comparing them to a reference atlas.

GraphST Clusters Annotated by SingleR

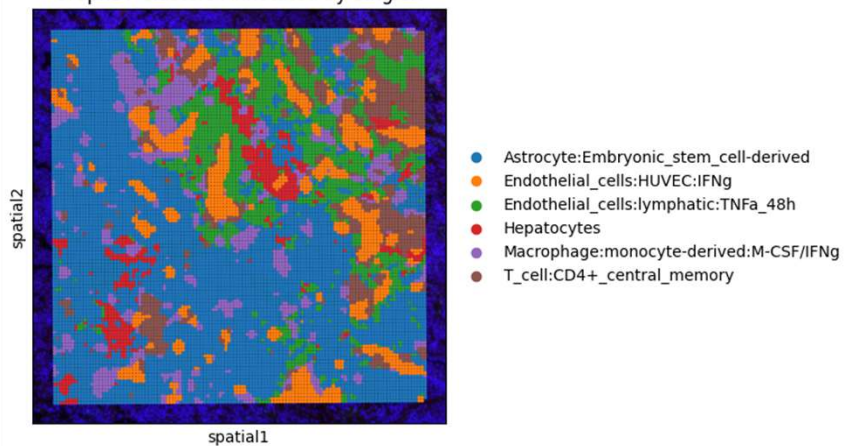


Fig: GraphSt Go Analysis

Banksy Clusters Annotated by SingleR

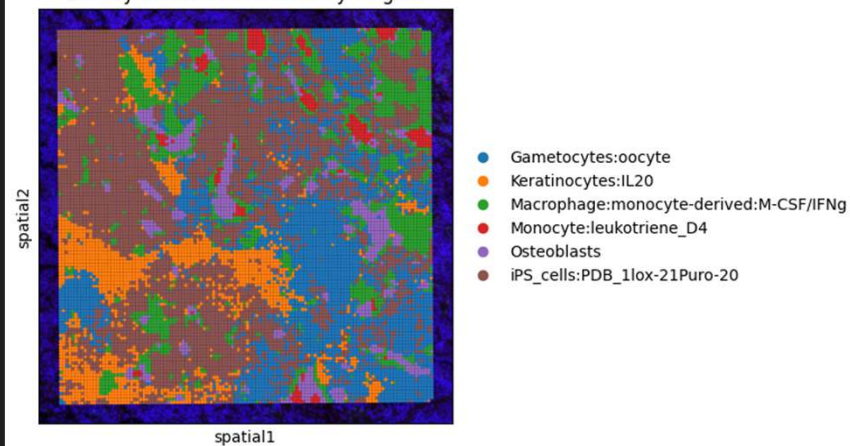


Fig: BANKSY SingleR Analysis

GraphST: Its transition-aware partitions often result in broader, **mixed identities** across gradients.

BANKSY: Its boundary-focused segmentation drives more **specialized cell-type calls** at tumor-stroma interfaces.

Key Finding 3: Deep Dive into CAF/TLS Niches

Mapping Key TME Niches: Cancer-Associated Fibroblasts (CAFs) and Tertiary Lymphoid Structures (TLS).

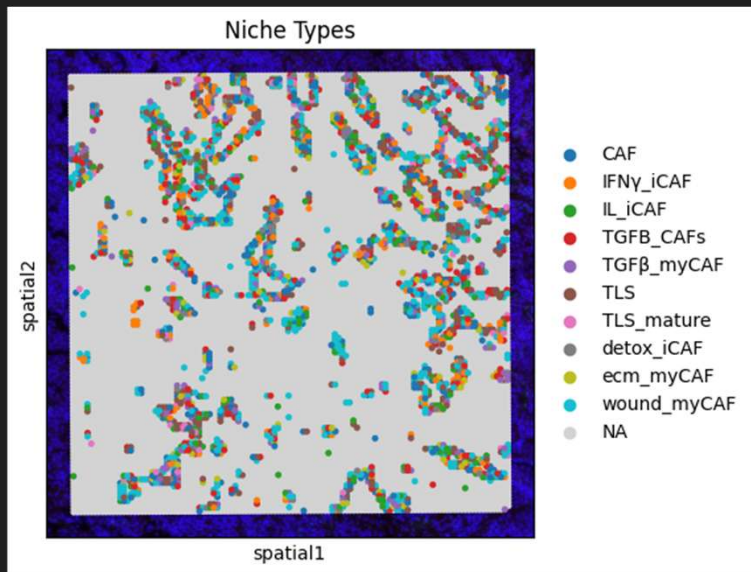


Fig: BANKSY SingleR Analysis

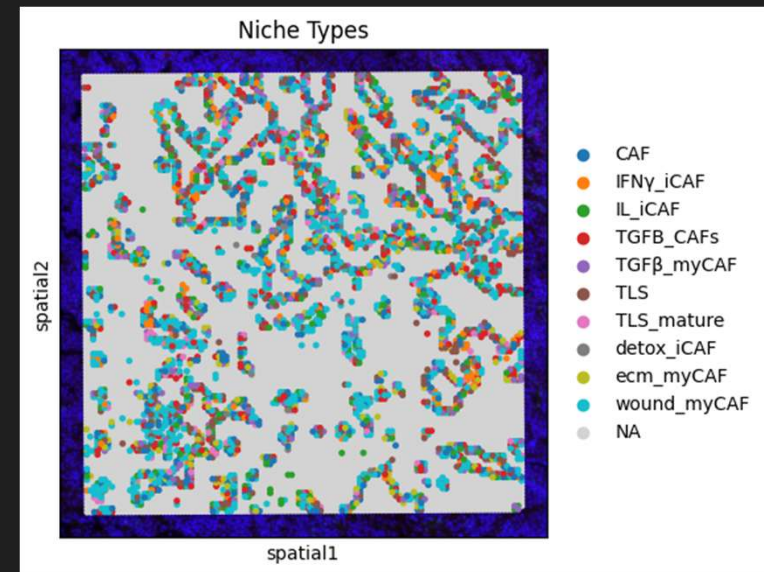


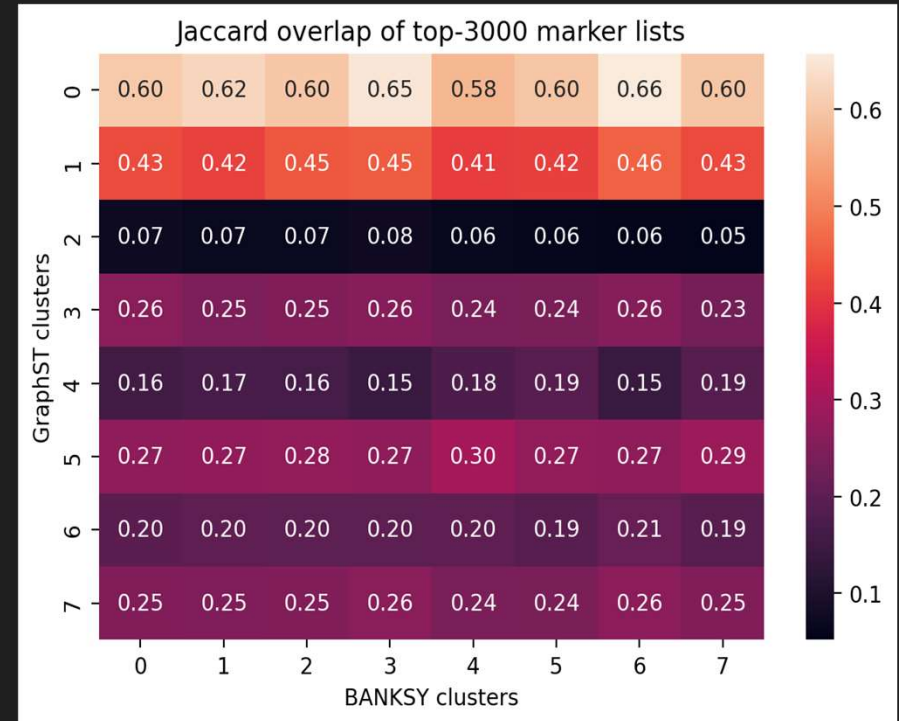
Fig: GraphST SingleR Analysis

BANKSY (Boundary-Focused): Identifies CAF programs as thin, crisp **rings** outlining tumor cores and TLS as compact **foci**.

GraphST (Transition-Focused): Identifies CAF programs as broader **belts** representing gradual transitions.

Quantifying the Agreement

Pair (G↔B)	Jaccard	Prec.	Rec.	F1	Fisher OR	p-value	Verdict
G0 ↔ B6	0.66	0.79	0.79	0.79	35.73	$<10^{-300}$	strong
G1 ↔ B2	0.45	0.62	0.62	0.62	7.34	$<10^{-300}$	strong
G2 ↔ B3	0.08	0.11	0.20	0.14	0.48	5.1×10^{-32}	anti-enriched
G3 ↔ B0	0.26	0.42	0.42	0.42	1.86	2.0×10^{-40}	moderate
G4 ↔ B5	0.19	0.32	0.32	0.32	1.02	7.0×10^{-1}	n.s.
G5 ↔ B4	0.30	0.47	0.47	0.47	2.57	1.5×10^{-91}	moderate
G6 ↔ B1	0.20	0.34	0.34	0.34	1.12	2.0×10^{-2}	borderline*
G7 ↔ B7	0.25	0.40	0.40	0.40	1.70	1.3×10^{-29}	moderate



- **Method:** Compared the top-3000 marker genes for each cluster pair using the Jaccard index and Fisher's exact test.
- **Strong Agreement:** The highest-matching pairs (Jaccard = 0.66 and 0.45) correspond to core tumor and immune niches.
- **Principled Disagreement:** Lower overlap pairs correspond to interface zones, where BANKSY creates finer niches while GraphST models a wider transition.

Conclusion

- GraphST and BANKSY are **complementary**, not competing, tools for spatial transcriptomics analysis.
- GraphST is ideal for studying global tissue architecture, broad domains, and smooth biological transitions.
- BANKSY excels at dissecting sharp micro-niche boundaries, localized gradients, and fine-grained spatial heterogeneity.

Future Work

- **Multi-scale Neighbourhood Analysis**

Develop and apply methods for analysing spatial neighborhoods at multiple resolutions to capture a more complete picture of tissue organization.

- **Broader Benchmarking**

Apply this comparative framework to a wider range of spatial transcriptomics datasets and tumour types to validate generalizability and identify context-specific strengths.

- **Histological Integration**

Integrate high-resolution histological images directly into the computational models for orthogonal validation and improved biological interpretability of identified spatial structures.

Thank You

Questions?